

21CSA528A

ESSENTIALS OF CYBERSECURITY

L-T-P-C: 2-0-1-3

Course Objectives

The student will learn to navigate a Linux CLI and run basic bash commands. Comfort in understanding the common architecture and platforms. Students will practice basic cryptography with code, as well as analyzing malicious binaries.

Course Outcomes

COs	Description
CO1	Understanding the usage and fundamental principles behind the GNU/Linux operating system
CO2	Grasping the essential principles behind currently used cryptographic algorithms
CO3	Learning reverse engineering skills associated with the x86/x86-64 platform
CO4	Understanding how to perform analysis of malicious binaries
CO5	Understand stack and heap exploits on system calls

CO-PO Mapping

PO/PSO							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	1	3	-	-	-	3
CO2	2	2	3	-	-	-	3
CO3	1	3	3	-	-	-	3
CO4	2	3	3	-	-	-	3
CO5	2	3	3	-	-	-	3
CAM	2	3	3	-	-	-	3

Syllabus

Unit I

Linux, Shell essential concepts, Shell I/O, Linux Paths and File Manipulation, Linux permissions,

Unit II

Cryptography, Kerkhoff's principle, Symmetric key encryption, MAC, Public key cryptography, Asymmetric key encryption, Digital signatures, TLS cryptography, Password hashing

Unit III

System security, System calls, Context switches, Malware Analysis, Static and dynamic analysis, Windows API, Malware identification, Malware disassembly

Textbooks / References:

1. Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2018). Operating systems: Three easy pieces. Arpaci-Dusseau Books LLC.
2. David Wong. (2021). Real-World Cryptography. Manning Publishing

AMRITA VISHWA VIDYAPEETHAM Amrita Online - Course Plan						
Week	Topic	Learning Objectives	eLearning Content	Video Lectures	Assessments	
1	Introduction to ECS	Introduction to Course	4 Videos	Introduction to Course		
		ECS Course Outline		ECS Course Outline		
		Introduction to Linux Terminal		Introduction to Linux Terminal		
		Linux File Abstraction		Linux File Abstraction		
2	Shell Essentials	Shell Essential Concepts	4 Videos	Shell Essential Concepts	Quiz-1	
		Shell Inputs/Outputs		Shell Inputs/Outputs		
		Linux Paths		Linux Paths		
		File Manipulation		File Manipulation		
3	Files and Directories	To get an overview on the different types of user types	3 Videos	Users and Groups	Assignment 1	
		To understand file permissions work		File Permissions		
		To understand directory permissions work		Directory Permissions		
4	Secure Communication	Understand the concept of files and file Organization	5 Videos	Securing Digitals Data		
		Understand the need of structured data representation and storage		Secure Communication		
		Give an overview on database. Understand the advantages and disadvantages of database with respect to files		Kerkhoff's Principle		Quiz-2
	Symmetric Key Algorithms	To get an overview on DBMS. Different data models and why data models are important.		Symmetric Key Algorithm		
				Virtual lab: https://cse29-iiith.vlabs.ac.in/exp/des/		
	Integrity & Authentication	Provide the concepts of Relational databases		Integrity & Authentication		
5	Message Authentication	Understanding Message Authentication	6 Videos	Message Authentication Codes	Assignment 2	
				Message Authentication using MAC		
				Symmetric vs Asymmetric Encryption		
	Symmetric & Asymmetric Encryption	Understanding Symmetric Encryption		Asymmetric Encryption		
		Understanding Asymmetric Encryption		Digital Signatures		
	Digital Signatures	Understanding Digital Signatures		Certificates		
	Certificates	Understanding Certificates		Virtual lab: https://cse29-iiith.vlabs.ac.in/exp/digital-signatures/		
6	Hybrid Encryption	Understanding Hybrid Encryption	6 Videos	Hybrid Encryption - I	Quiz 3	
	Diffie-Hellman	Learning about forward secrecy		Hybrid Encryption - II		
	Transport Layer Security	Diffie-Hellman		Forward Secrecy		
		Understanding transport layer security		Diffie-Hellman		
	TLS Authentication	Learning about TLS Authentication		Transport Layer Security		
				TLS Authentication		

	7	Different Authentication Methods	Understanding RSA	8 Videos	TLS Authentication with RSA	Assignment 3
			Understanding key exchange		TLS Key Exchange	
			Understanding symmetric key encryption.		TLS Symmetric Encryption-I	
			Understanding the concept of padding		IV & Padding	
			Cryptographic Hashes		TLS Symmetric Encryption-II	
	8	Keys and Passwords	General recommendations and Keys & passwords	7 Videos	TLS Symmetric Encryption-III	Quiz 4
					Cryptographic Hashes	
					TLS Conclusion	
					Virtual Lab: https://cse29-iiith.vlabs.ac.in/exp/hash-functions/	
					Key & Passwords	
	9	System Security	Familiarizing system security, OS, and processes	7 Videos	General Recommendations - I	Quiz 4
					General Recommendations - II	
					Introduction to System Security	
					OS	
					Processes	
	10	Assembly Programming	Introduction to context switching, stacks	6 Videos	System Calls	Assignment 4
					Introduction	
					Context Switch	
					Stack	
					CPU Programming	
	11	Virtual Memory	Introduction to assembly programming	5 Videos	Assembly Programming I	Quiz 5
					Assembly Programming II	
					Assembly Programming III	
					Assembly Programming IV	
					Endianness	
	12	Basics of LLB	Familiarizing with Virtual Memory	5 Videos	Virtual Memory	Quiz 6
					Segmentation - I	
					Segmentation II	
					Paging	
					TLB	
	13	Libraries	Understanding Libraries	4 Videos	Multilevel Paging	Quiz 6
					Swapping	
					Libraries I	
					Libraries II	
					Libraries III	
	14	How does swapping works	Introduction to Malware Analysis	4 Videos	Libraries IV	Quiz 6
					Introduction to Malware Analysis	
					Static Analysis - I	

		Introduction to Malware Analysis	Introduction to Static Analysis		Static Analysis - II Static Analysis - III	
	14	Malware Analysis	Types of malware analysis	4 Videos	Static Analysis – IV Dynamic Analysis – I Dynamic Analysis - II Malware Analysis Conclusion	Quiz 7

Virtual lab link for reference

- [Breaking the Shift Cipher](#)
- [Breaking the Mono-alphabetic Substitution Cipher](#)
- [One-Time Pad and Perfect Secrecy](#)
- [Message Authentication Codes](#)
- [Cryptographic Hash Functions and Applications](#)
- [Symmetric Key Encryption Standards \(DES\)](#)
- [Symmetric Key Encryption Standards \(AES\)](#)
- [Diffie-Hellman Key Establishment](#)
- [Public-Key Cryptosystems \(PKCSv1.5\)](#)
- [Digital Signatures](#)

Evaluation Policy

Internal	Quiz- All the 7 quizzes are mandatory	30%	100 %
External	Section 1-End semester MCQ (Unidirectional) – 50Marks	70%	
	Section 2 - End Semester Descriptive – 20Marks		

Note: Assignments -4 no's (Ungraded)

Faculty Information

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